LAP-TEQ PLUS — User Manual

TEQSAS PRODUCTS

LAP-TEQ PLUS

User Manual

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1 Before you start

Intended use

The LAP-TEQ PLUS is used to measure the inclination of loudspeaker systems.

The LAP-TEQ PLUS measures the vertical angle between its longitudinal axis and the surface of the earth. This provides information about the vertical alignment of the loudspeaker element.

Any other use that is not described in these instructions for use is unauthorised misuse. The manufacturer is not liable for any damage resulting from this.

Never use the device as a display device or in public traffic (road traffic, aviation, etc.).

What do the symbols used mean?

Hazard warnings and notes are clearly marked in the instructions for use. The following symbols are used:

DANGER

Immediate danger to life or injury!

Immediately dangerous situation that will result in death or serious injury.

WARNING

Risk of death or injury!

Generally dangerous situation that may result in death or serious injury.

CAUTION

Potential risk of injury!

Dangerous situation that may result in injury.

ATTENTION

Risk of damage to the appliance!

Situation that may result in material damage.

> NOTE
>
> Information provided for a better understanding of the processes.

2 For your safety

WARNING!

Failure to observe the safety information and instructions may result in electric shock, fire and/or serious injury!

Read all safety information and instructions.

General safety instructions

- Keep all safety information and instructions for future reference. The term “appliance” used in the safety instructions refers to mains-powered appliances (with mains cable) and battery-powered appliances (without mains cable).
- Danger to life and accidents for infants and children! Never leave children unattended with packaging material and product. There is a risk of suffocation from packaging material and a risk to life from strangulation. Children often underestimate the dangers. Always keep children away from the product. The product is not a toy.
- To ensure safe handling of this appliance, the user must have read and understood these instructions for use before using the appliance for the first time.
- If you sell or pass on the appliance, it is essential that you also hand over these instructions for use.
- This appliance is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and/or knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety.
- The appliance may only be used if it is in perfect working order and fully assembled. If the appliance or any part of it is defective, it must be taken out of service and repaired or disposed of properly.
- Use the appliance only for the purpose for which it is intended.
- Keep children away from the appliance! Keep the device safe from children and unauthorised persons.
- Only use and store the device within the permissible ambient conditions (temperatures, humidity, etc.).

Handling laser beams

- Do not look into the beam, even from a great distance.
- Never aim the measuring beam at people, other living beings or reflective surfaces.
- Standard laser goggles do not protect against the dangers of laser radiation. They merely make the laser beam easier to see.

Workplace safety

- Keep your work area clean and well lit. Untidiness or unlit work areas can lead to accidents. Observe the workplace and accident prevention regulations applicable in your country.
- Do not work with the electrical appliance in potentially explosive atmospheres containing flammable liquids, gases or dust. Power tools generate sparks that can ignite the dust or vapours.

Electrical safety

- Static electricity can occur in dry environments. In dry rooms, touch a metal object to discharge the static electricity before operating the appliance.
- Do not misuse the cable to carry or hang up the appliance or to pull the plug out of the socket. Keep cables away from heat, oil, sharp edges or moving parts of the appliance. Damaged or tangled cables increase the risk of damage to the appliance.
- If you use the appliance outdoors, only use extension cables that are also suitable for outdoor use. The use of an extension cable suitable for outdoor use reduces the risk of damage to the appliance.

Safety of persons

- Be attentive, pay attention to what you are doing and use common sense when working. Do not use the appliance when you are tired or under the influence of drugs, alcohol or medication. A moment of inattention during use can lead to serious injury. The INCLINOMETER must be securely screwed to the flying frame or secured against falling using a suitable safeguard.

Handling battery-powered devices

- Only chargers approved by the manufacturer with the values specified on the battery's nameplate may be used. Using non-approved chargers may cause injury and material damage due to exploding batteries.
- Only use the batteries intended for use with this device. The use of other batteries can lead to injury and risk of fire.
- Do not expose the battery to heat, fire, water or moisture. There is a risk of explosion.
- If the battery is damaged or used improperly, vapours may escape. Ensure good ventilation and consult a doctor if you experience any symptoms. The vapours may irritate your respiratory tract and cause illness.

Safety information for chargers

- Keep the charger away from rain or moisture. Water entering a charger increases the risk of electric shock.
- Keep the charger clean. Contamination poses a risk of electric shock.
- Check the charger, cable and plug before each use. Do not use the charger if you discover any damage.

- Do not open the charger yourself and only have it repaired by qualified specialists and only with original spare parts. Damaged chargers, cables and plugs increase the risk of electric shock.
- Do not operate the charger on an easily flammable surface (e.g. paper, textiles, etc.) or in a flammable environment. There is a risk of fire due to the heat generated during charging.
- Supervise children and ensure that they do not play with the charger. Children and persons with mental or physical impairments may only use the charger under supervision or if they have been instructed in its use. Careful instruction reduces incorrect operation and injuries.

Service

- Only have your appliance repaired by qualified specialists and only with original spare parts. This will ensure that the safety of the power tool is maintained.
- Ensure adequate lighting when working with the appliance. Poor visibility can lead to an increased risk of accidents.
- Do not use any accessories that have not been specially designed and recommended by the manufacturer for this appliance. Just because you can connect the accessories to your appliance does not guarantee trouble-free and safe use.
- The device may show signs of wear and tear over time.
- The use of unauthorised accessories may cause damage or wear that is not covered by the warranty.

Special device instructions

- Do not use the appliance in potentially explosive atmospheres or in the vicinity of flammable liquids or gases!
- Paint and stickers can stick to moving parts and impair proper operation.
- If you have an allergic reaction to the colour or metallic parts of the appliance, it may cause itching, eczema or swelling of the skin. If this is the case, stop using the appliance and consult a doctor.
- Broken glass or plastic can cause injury. Have the appliance repaired at the service centre.
- Do not drop the device and do not subject it to shocks.
- The device or parts of it may be damaged if dropped, bent or deformed.
- Magnetic fields can cause the device to malfunction or the battery to discharge.
- The device has safety precautions that limit the leaked output of the laser beam: no changes may be made to the optical system.
- This device is classified as a Class 3R laser product in accordance with the international standard IEC 60825. It works with a visible laser beam that is harmless under normal operating conditions.

- The system has a Class 3R laser. This laser class may only be operated by commercial users. The operator is liable for personal injury and material damage caused by operation.
- Symbols on your device must not be removed or covered. Labels on the appliance that are no longer legible must be replaced immediately.

NOTE

Read and observe the instructions for use before commissioning.

RISK OF EYE DAMAGE!

Never look directly into the laser beam.

3 Product description

3.1 Scope of delivery

- 1× LAP-TEQ Display PLUS (display module for short)
- 2× LAP-TEQ PLUS INCLINOMETER
- 1× USB charging cable
- 1× Calibration cable
- 1× Safety instructions
- 1× Transport case

3.2 Your device at a glance

1. LAP-TEQ Display PLUS
2. Display
3. Power button
4. Calibration button
5. Charge level indicator
6. USB charging socket
7. XLR cable socket
8. LAP-TEQ PLUS INCLINOMETER
9. Pressure compensation membrane opening
10. Laser outlet
11. Base plate with mounting plate
12. XLR cable socket
13. Calibration cable

A complete LAP-TEQ PLUS angle measuring system consists of one display module and at least one INCLINOMETER.

The INCLINOMETER is intended for mounting, e.g. on the topmost element of a line array or on the bumper, and measures the vertical angle between its longitudinal axis and the surface of the earth. This provides information about the vertical alignment of the loudspeaker element or the entire array.

The laser beam of the INCLINOMETER serves as visual feedback for the area being supplied with sound. This way you can check whether the height of the loudspeaker system is set optimally to cover, for example, an audience tier or a balcony.

If the laser beam hits a projection surface at close range, a square frame and several dots in line with the cross axes become visible in addition to the actual laser cross. These artefacts are no longer visible at the typical working distances of the system and do not constitute a defect.

The products LAP-TEQ Display, LAP-TEQ Sensor, LAP-TEQ Display PLUS and LAP-TEQ PLUS INCLINOMETER are compatible with each other and can be operated in any combination.

NOTE

The LAP-TEQ Sensor is restricted to operation with the base plate facing down.

Calibration of an INCLINOMETER from the display module is only possible with the combination of LAP-TEQ Display PLUS and LAP-TEQ PLUS INCLINOMETER.

To calibrate a LAP-TEQ PLUS INCLINOMETER without a LAP-TEQ Display PLUS, please contact your sales partner.

Compatibility overview

	Sensor – Operation	Sensor – Calibration	INCLINOMETER – Operation	INCLINOMETER – Calibration
Display	Yes	Internal	No	No[1]
Display PLUS	Yes	Internal	Yes	Yes

[1] Contact service partner.

4 Powering up

4.1 Charging the battery

The battery must be charged before initial use.

If the device is not connected to a power source, the battery charge level is shown on the display when the device is switched on.

> ATTENTION!
>
> Risk of damage to the appliance!
>
> Do not deep-discharge the battery.

> NOTE
>
> The Li-Polymer battery can be charged at any time without shortening its service life. Interrupting the charging process does not damage the battery.

> NOTE
>
> If the device is not used for a long time, it discharges. In that case it must be charged before use.
>
> Please note that the battery only reaches its full capacity after several full discharge and charge cycles.

> ATTENTION!
>
> Risk of damage to the appliance!
>
> Forcefully or incorrectly connecting the charging cable can damage the multifunction socket or other parts of the device. Connect the cable to the device with the correct side facing the socket.

- Plug the USB charging cable (1) into the USB socket (2) on the display module.
- Connect the other end of the USB charging cable to a suitable USB power source (e.g. USB port of a computer or a USB power adapter).

When the device is switched on, the charging process is indicated by a blue arrow on the display.

The charge level is shown by the indicator light (4):

- Red LED – charging
- Green LED – fully charged
- End the charging process once the battery is charged.
- Disconnect the USB charging cable from the power source as long as you are not using it.

4.2 Connection

> ATTENTION!
>
> Risk of damage to the appliance!
>
> Moisture and liquid can cause damage to parts or electronic circuits inside the device.
>
> - Do not switch on the device when it is damp.
> - If the device is already switched on, switch it off.
> - If the device cannot be switched off, leave it in this state to dry.
> - Wipe the device dry with a dry cloth and bring it to a service centre.

> ATTENTION!
>
> Risk of damage to the appliance!
>
> Do not insert any pointed objects into the opening of the pressure compensation membrane of the INCLINOMETER.

> ATTENTION!
>
> Risk of damage to the appliance!
>
> All components are maintenance-free. Never open the devices, otherwise the warranty will be invalidated.

- Connect the display module (1) and an INCLINOMETER (3) using a standard 3-pin XLR cable (2).

Any number of INCLINOMETER units can be connected to a single display module in succession. For example, three INCLINOMETER units can be used to verify correct alignment of the left and right PA sides as well as a centre cluster. Alternatively, an INCLINOMETER can be permanently attached to each bumper.

For correct operation, the display module must be connected to an INCLINOMETER via a standard XLR connection cable. The INCLINOMETER units are powered via the XLR cable. There are no rechargeable or non-rechargeable batteries inside the INCLINOMETER.

If the cable between display and INCLINOMETER is longer than 75 m, the brightness of the laser is reduced. The measurement function still works.

If you want to bridge distances greater than 75 m, an XLR cable with a cross-section larger than 0.22 mm^2 must be used. This reduces the voltage drop along the cable.

> NOTE
>
> The supplied calibration cable can also be used for a function test. Using the calibration cable alone does not trigger the calibration process.

- When mounting the INCLINOMETER units, use the flange holes (1) on the bottom of the housing.

For secure mounting on a bumper without a dedicated LAP-TEQ mount, the optional TEQSAS LAP-TEQ magnetic holder is available. The INCLINOMETER must be securely screwed to the flying frame or secured against falling using a suitable safeguard.

4.3 Initial commissioning

The INCLINOMETER units are calibrated at the factory so that the base plate must face down. If you want to use the INCLINOMETER in another orientation, it must be recalibrated. Four positions are possible (90° increments around the longitudinal axis), see Chapter 8 Calibration.

For initial commissioning, the base plate must be positioned facing down.

5 Usage

5.1 Switching on/off

Check that the device is in a safe condition:

- Check whether there are any visible defects.
- Check that all parts of the device are firmly attached.

> WARNING!
>
> Risk of eye injuries from laser beams!
>
> If the laser beam enters the eyes through optical instruments, irreparable damage may occur.
>
> Never look into the laser or aim the laser beam at reflective surfaces!

- Make sure that no persons are in the effective range of the laser.
- Switching on: press the “Power button” for approx. 3 seconds.

If the system has been put into operation correctly, the green laser lights up and the current angle reading is shown on the display.

The angle reading is updated every 2 seconds. This integration over time is necessary to avoid constant fluctuation of the last digit, as would be caused by wind movement, for example.

> NOTE
>
> If the “CAL button” on the display module is pressed during operation, the INCLINOMETER is not put into calibration mode. Instead, brief calibration instructions appear on the display. After a few seconds, the display automatically returns to the angle reading.

- Switching off: press the “Power button” again.

5.2 Display indications

If no cable, a defective cable or no INCLINOMETER is connected to the display module, the warning “no sensor” appears on the display when the display module is switched on. Please establish a correct connection between the display module and the INCLINOMETER.

If the text “over” appears on the display, the permissible measurement range of +/– 50° has been exceeded. In practical use, this measurement range is sufficient.

If three dashes appear on the display, the INCLINOMETER is being used in an orientation in which it has not been calibrated.

After switching off, the last measured value is written to the internal memory of the display module. When the device is switched back on, this value appears on the lower line of the display. This makes it easier to compare the horizontal alignment of the left and right PA sides for identical readings.

5.3 INCLINOMETER Setup

Open INCLINOMETER Setup:

- Connect the INCLINOMETER to the display module using the calibration cable.
- Switching on: press the “Power button” for approx. 3 seconds.
- Wait until a measured value appears on the display.
- Press and hold the “CAL button” on the display module until “las mode” appears on the display.

INCLINOMETER Setup is now open.

The following diagram shows the navigation through all menu items and settings.

6 Cleaning and care

DANGER!

Danger to life from electric shock!

Disconnect the mains plug from the socket before carrying out any work on the device.

ATTENTION!

Risk of damage to the appliance!

Do not immerse the device in water when cleaning, as water can penetrate the electrical system and damage it.

ATTENTION!

Possible product damage!

The surface can be damaged. Do not use scratching or abrasive objects or aggressive cleaning agents for cleaning.

- Use a soft, lint-free cleaning cloth dampened with mild soapy water to clean the housing.
- Avoid scrubbing motions on the display window and on the glass window covering the laser outlet.
- Do not use chemicals, cleaning agents or solvents. This can lead to discolouration and corrosion of the outside of the device or result in an electric shock or fire.

7 Issues and Help

CAUTION!

Risk of injury!

Improper repairs can result in your appliance no longer functioning safely. You are putting yourself and your surroundings at risk.

It is often only minor faults that lead to a malfunction. In most cases, you can easily rectify these yourself. Please refer to the following table before contacting your dealer. This will save you a lot of trouble and possibly also costs.

If the device requires servicing, take it to your sales partner. The battery built into this product cannot be replaced by the user. Please contact our service centre for information on replacement.

Error/Fault	Cause	Remedy
Device does not work	Device defective?	Contact service.
	Battery discharged?	Charge the battery.
	Battery defective?	Contact service.
Laser not visible	Poor ambient conditions?	See note on ambient conditions in this manual.
	Cable too long?	Use an XLR cable with a larger cross-section.
	Measurement distance too long?	If possible, reduce the measurement distance.
“over” appears on the display	—	Permissible measurement range of +/– 50° has been exceeded.
Three dashes appear on the display	—	Recalibrate the device.

If you cannot rectify the fault yourself, please contact our service team directly. Please note that improper repairs will also invalidate the warranty and you may incur additional costs.

7.1 Service centre

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TEQSAS GmbH
Otto-Hahn-Straße 20a
50354 Hürth
Germany
Tel.: +49 (0)2233 611-500
E-Mail: service@teqsas.de
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7.2 Observe ambient conditions

The visibility of the laser beam depends on the ambient brightness. As a general rule:

- Subdued light – good visibility
- Direct sunlight – poor visibility
- Rain, dust or steam – poor visibility

Other adverse influences:

- Measurements through glass or plastic panes
- Contaminated laser outlet
- Strong temperature fluctuations: always allow the device to acclimatise in the transport case for some time before use in very cold or warm environments.

8 Calibration

> NOTE
>
> The INCLINOMETER is calibrated at the factory for operation with the base plate facing down.
>
> If you wish to operate the INCLINOMETER with another orientation of the base plate (up, left or right), calibration in the desired position is required.
>
> Once all positions have been fully calibrated, the INCLINOMETER can be used in all positions without further calibration.

A new calibration may become necessary if the INCLINOMETER has been exposed to extreme shocks (e.g. from being dropped). Calibration is required if the INCLINOMETER is not to be operated with the base plate facing down — i.e. with the base plate facing up, left or right. The calibration procedure is identical for all positions except for the orientation of the base plate, and is therefore described in this manual using the variant “base plate facing down” as an example.

In addition to the INCLINOMETER to be calibrated, the following materials are required:

- Display module
- Calibration cable (see image in Chapter 3.2). Standard XLR cables are not suitable.
- Spirit level
- Set square

Step 1
- Connect the INCLINOMETER to the display module using the calibration cable.

Step 2
- Check that the reference surface on which calibration is to be carried out is level.
- Position the INCLINOMETER with the base plate on the reference surface. The laser outlet faces left.

Step 3
- Press and hold the “CAL button” on the display module.
- Press the “Power button” on the display module.
- Once the “Welcome” text appears at the bottom of the display, release both buttons.

The system is now in calibration mode and indicates this in the upper area of the display by showing “start calibrte?”.

> NOTE
>
> If the “CAL button” on the display module is pressed during operation, the INCLINOMETER is not put into calibration mode. Instead, brief calibration instructions appear on the display. After a few seconds, the display automatically returns to the angle reading.

- Press the “CAL button” again to start the actual calibration.

The display shows the question “calibrte 0.0° A?”.

- Press the “CAL button” again to start calibration in the “A” direction.

If the display shows “Sens pos incorrect”, please check the orientation of the base plate.

If the procedure is correct, the display shows “don’t move” – followed by a numerical value for “A” and the question “calibrte 0.0° B?”.

Step 4

- Rotate the INCLINOMETER so that the laser outlet faces right.
- Press the “CAL button” again to start calibration in the “B” direction.

The display again shows “don’t move” – followed by a numerical value for “B” and the question “calibrte +45.0°?”.

Step 5

- Use a set square to align the INCLINOMETER so that the laser shines upwards at a 45° angle.
- Press the “CAL button” again to start calibration at “+45.0°”.

The display shows “don’t move” – followed by a numerical value for “+45.0°” and the question “calibrte –45.0°?”.

Step 6

- Use a set square to align the INCLINOMETER so that the laser shines downwards at a 45° angle.
- Press the “CAL button” to start calibration at “–45.0°”.

The display shows “don’t move” – followed by a numerical value for “–45.0°”.

Calibration for the orientation with the base plate facing down is now complete. The display shows “Calb Data stored”.

Step 7

If the INCLINOMETER is operated exclusively with the base plate facing down, nothing further needs to be done.

If operation with another orientation is desired, the steps from Step 8 onwards must be performed.

To leave calibration mode, switch off the display module by pressing the “Power button”. The next time it is switched on – without pressing the “CAL button” – the device returns to its normal operating mode.

Step 8
If you want to operate the INCLINOMETER with the base plate in another position, calibration must be continued.

The order of the four possible positions is fixed as follows (viewed from the XLR connector of the INCLINOMETER):

Orientation	Display text
I. Base plate down (factory calibration)	Mpl Down[1]
II. Base plate up	Mpl Up
III. Base plate left	Mpl Left
IV. Base plate right	Mpl Right

[1] Mpl: Mounting plate.

> NOTE
>
> For better clarity as to which orientation (I.–IV.) is currently being calibrated, the display briefly shows the corresponding display text from the table.

> NOTE
>
> Note on Step 3: as the system is already in calibration mode, the “Power button” does not need to be pressed. Calibration starts immediately with calibration of position “A” when the “CAL button” is pressed.

If Step 2 to Step 6 have been performed correctly with the changed orientation of the base plate without the display showing “Sens pos incorrect” in between, calibration of this orientation finishes with the display showing “Calb Data stored”.

The INCLINOMETER can now be operated with the base plate facing down and the base plate facing up. If these positions are sufficient, calibration mode can be exited by switching off using the “Power button”.

If the lateral positions are also desired, Step 2 to Step 6 must first be repeated with the base plate facing left (viewed from the XLR connector of the INCLINOMETER). For the orientation with the base plate facing right (viewed from the XLR connector of the INCLINOMETER), calibration Step 2 to Step 6 must be performed once more.

The INCLINOMETER can now be operated in all four positions.

8.1 Calibration of 2 axes

- Press and hold the “CAL button” on the display module.
- Press the “Power button” on the display module.
- Once the “Welcome” text appears at the bottom of the display, release both buttons.

The system is now in calibration mode and indicates this in the upper area of the display by showing “start calibrte?”.

- Press and hold the “CAL button” on the display module for approx. 20 seconds to enter “2-axis mode”.

The display shows the question “calibrte 0.0° A?”.

- For further calibration, follow the description as outlined in Step 3, point 5 up to and including Step 6.

In addition, calibration of the X-axis at +45°/–45° follows.

After completing calibration, the X- and Y-axis values appear on the display. After switching off and back on, the “spirit level” view is shown.

9 Storage

- Clean and dry all parts and store them in a clean and dry place.
- If you do not use the device for a long time or are transporting it, use the original packaging.

ATTENTION!

Risk of damage to the appliance!

High temperatures can cause the display to malfunction, damage the device or cause the battery to explode.

Do not store the device in very hot places such as the interior of a car in summer or places exposed to direct sunlight.

10 Disposal

The packaging is made of environmentally friendly materials that can be disposed of at local recycling centres.

Devices marked with the symbol shown alongside must not be disposed of with household waste. You are obliged to dispose of waste electrical and electronic equipment separately.

You can find out how to dispose of old appliances from your local authority.

By disposing of old appliances separately, you are sending them for recycling or other forms of reuse. In doing so, you will help to prevent harmful substances from being released into the environment.

10.1 Disposing of batteries

Batteries that are no longer needed do not belong in household waste — they must be disposed of in accordance with regulations.

- Take used batteries to a battery collection point at the dealer or municipality.
- Do not burn batteries. Risk of explosion!
- Do not burn rechargeable batteries. Risk of explosion!

10.2 Removing the battery

> NOTE
>
> Before disposing of the device, the battery must be removed. If the battery is defective, it can be replaced by a service workshop. Please contact your sales partner.

- Remove the cover caps on the display module using a small screwdriver.
- Unscrew the screws (1).
- Disconnect the cable (2).
- Remove the battery (3) and dispose of it properly.

11 Technical Data

LAP-TEQ Set

LAP-TEQ Set	
Angle measurement resolution	0.1°
Permissible ambient temperature for use	0 – 40 °C
Permissible ambient temperature for storage	–20 – 50 °C
Total weight of the set	1.485 kg
Transport case	293 × 228 × 80 mm

LAP-TEQ Display PLUS

LAP-TEQ Display PLUS	
Protection rating	IP53
Display	1.8″ TFT display
Dimensions	138.5 × 78 × 38.5 mm
Weight	434 g

LAP-TEQ PLUS INCLINOMETER

LAP-TEQ PLUS INCLINOMETER	
Protection rating	IP54
Optics	5° cross optics (from Q3/2018)7° cross optics (until Q2/2018)
Laser class of laser diode	3R
Laser class of laser system	2M
Laser colour	520 nm (green)
Dimensions	121.5 × 38.5 × 32.7 mm
Weight	170 g

Battery

Battery	
Battery type	Li-Polymer
Nominal voltage	3.7 V
Capacity	2000 mAh
Charging time	4–5 h

Accessories

Accessories	
Calibration cable	XLR 3-pin, 500 mm
USB charging cable	Micro USB (APCBU10BBECSTD)

The manufacturer reserves the right to make technical changes to the product / product group without prior notice.

EU Declaration of Conformity

Figure 1: CE

in accordance with EU Directive 2014/30/EU (Electromagnetic Compatibility) of 26 February 2014 and 2014/35/EU (Low Voltage Directive) of 26 February 2014.

We hereby declare that the device designated below, in its design and construction and in the version placed on the market by us, complies with the essential health and safety requirements of EU Directive 2014/30/EU and the requirements for electromagnetic compatibility. This declaration shall lose its validity if the device is modified without our agreement.

Manufacturer

TEQSAS GmbH
Otto-Hahn-Straße 20a
50354 Hürth

The manufacturer bears sole responsibility for issuing this declaration of conformity.

Description of the device:
LAP-TEQ PLUS

and its identical or similar derivatives.

Place and date of issue: Hürth, 16 November 2018